



Simulation of LiDAR Under Fog with Generative Adversarial Networks for Robust 3D Object Detection

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Abstract

Collecting real LiDAR data from actual scenarios in adverse weather is an expensive and time-consuming process, yet crucial for developing robust perception algorithms. Publicly available datasets published for relevant purposes are often limited to acquisitions under clear weather, which leads to a lack of diversity for the most challenging and unexpected situations that can be encountered in the real world. Virtual datasets that are generated to include emulated adverse weather conditions will significantly facilitate the development and testing processes of more robust perception algorithms. In this paper, we propose a Wasserstein Generative Adversarial Network (WGAN)–Long Short-Term Memory (LSTM) network architecture trained on real continuous LiDAR point clouds gathered in a foggy environment. The output of the generator portion of the network is projected onto a fully simulated virtual scenario. Instead of directly addressing the domain gap induced by this method, we show indirectly that training on the generated dataset improves the performance and robustness of 3D object detection algorithms.

Keywords Generative adversarial network · LiDAR · Adverse weather · 3D object detection · CarMaker

Abbreviations

LiDAR Light Detection and Ranging
GAN Generative Adversarial Network
LSTM Long Short-Term Memory

List of Symbols

G Generator
 D Critic or discriminator
 $\mathbb{P}_g$ Generated data distribution
 $\mathbb{P}_r$ Real data distribution
 z Random noise input to generator

1 Introduction

The development and deployment of autonomous vehicles introduce a variety of concerns, particularly surrounding their reliability and passenger safety. These concerns include technical issues that primarily stem from the opaque nature of computer vision perception algorithms and their inherent reliance on training data derived from physical sensors. As radar, LiDAR, and camera sensors that utilize the properties of electromagnetic waves to produce data that implicitly represent the surrounding environment of the originating vehicle have become dominant, efforts to achieve high-level semantic analysis, such as 3D object detection using common datasets provided by validated sources, have become standard practice.

Although methods that use LiDAR are generally among the more performant, the vulnerability of LiDAR to adverse weather conditions—including rain, snow, and fog—poses a significant limitation (Bijelic et al., 2018; Heinzler et al., 2020). As atmospheric particles can interfere with the light rays emitted from a LiDAR sensor, adverse weather eventually leads to erroneous or obfuscated representations of the surrounding environment, impacting the performance of subsequent perception algorithms (Goodin et al., 2019; Rasshofer et al., 2011). Given that data-driven methods currently constitute the majority of approaches for processing

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LiDAR point clouds, this leads to a major caveat that must be addressed appropriately.

To ensure and enhance the robustness of perception, exposure to diverse scenarios and environments during neural network training is crucial. However, most studies utilize automotive datasets from external publications (Caesar et al., 2020; Geiger et al., 2012; Houston et al., 2020; Pham et al., 2019; Sun et al., 2020), such as KITTI and nuScenes, which come with a few inevitable assumptions—identicality and consistency of sensor models, vehicle models, and seasonal atmospheric characteristics. This lack of flexibility in data acquisition setups not only hinders the pursuit of a universal solution applicable to a wide range of vehicles and sensors but also impedes the development of solutions that are more resilient to varying weather conditions and are robust against external interferences. We aim to approach this problem with a neural network powered method to inject weather effects on simulated LiDAR point cloud data.

In recent years, machine learning that utilizes the form of a neural network has become a widely adopted practice for many modeling tasks in multiple domains, including science and engineering. The insight in neural networks is that the sheer amount of data provided to the model combined with multiple hidden layers that form a high-dimensional nonlinear equation can help catch the minute details of an unknown model just by gravitating toward a desirable state (Rumelhart et al., 1986). Generative modeling is no exception, with the advent of Generative Adversarial Networks (GANs)—a type of unsupervised learning neural network known for generating visually plausible images using competing pairs of neural networks.

One strong alternative to GAN is the diffusion-based model, which has recently emerged as a powerful generative method capable of producing high-quality outputs. However, their iterative denoising process often results in longer inference times, making them less suitable for real-time simulation scenarios. In contrast, GANs offer the advantage of a single forward pass through the generator network, ensuring rapid and predictable inference performance essential for our real-time LiDAR fog simulation.

To address the spatial distribution of fog detections, the same principle of convolutional processing on images is applied to LiDAR cloud points laid out onto a 2D array. We then extend the region of interest to include temporal information by integrating with a recurrent neural network to capture the time-variant aspect of LiDAR in addition to the spatial information understood by convolution operations. To further enhance the capability of the neural network architecture, Long Short-Term Memory (LSTM) is used for the RNN (Elman, 1990) component, and Wasserstein GAN with Gradient Penalty (WGAN-GP) is used for the GAN component. LSTM addresses the exploding/vanishing

gradient problems that can arise with long data sequences, while WGAN improves convergence and stability using the Earth Mover's distance (Wasserstein distance) instead of the Jensen–Shannon divergence derived from binary cross entropy. Also, the use of gradient penalty enforces Lipschitz continuity, which is crucial for the stability of the training process.

Thus, our selection of GANs over diffusion models is not only justified by the necessity of real-time operation and predictable inference latency, but also supplemented by techniques effectively to address their known drawbacks through architectural adaptations.

For the validation of our methodology for closing the domain gap, we plan to test our approach using one of the vehicle development procedure archetypes: Vehicle-in-the-loop (VIL). VIL is a variant of the “X-in-the-loop” testing methodologies, where “X” represents different testing conditions and requirements. Despite the variations, all XIL modes share a common hurdle that they intend to overcome: vehicle testing is expensive and time-consuming. This challenge that motivates our use of VIL is identical to the one that motivates LiDAR simulation enhanced with weather artifacts.

In VIL simulation, we overlay the simulated environment onto an actual vehicle while reflecting the real vehicle's dynamic status onto its digital twin. Virtual interactions are simulated using a commercial solver, which is then recorded with a virtual LiDAR sensor that utilizes ray tracing in a GPU, producing a formatted dataset ready for training. By running the sensor model on a separate piece of hardware while the main solver runs on a Real-Time Operating System (RTOS), we ensure the real-time condition necessary for the VIL simulation.

Since virtual scenarios are used for VIL tests, we plan to acquire simulated LiDAR point clouds from two variants of the scenarios: one with fog and one without fog. By comparing the training results on these two variants with well-known published 3D object detection architectures, we aim to demonstrate the feasibility of our method in improving the robustness of perception algorithms.

In short, we make the following contributions. First, we present a neural network architecture that generates fog-like artifacts on the fly in real time. The generated fog not only follows the spatial distribution but also the temporal distribution of natural fog to emulate the relative movement between the body of fog and the vehicle. Second, we present a methodology for systematical pipelining of data across different phases, including acquisition, training, generation, and simulation. Third, by following the suggested methodology, we do not depend on pre-acquired datasets, but instead acquire our own dataset and train our model for that dataset, assuring consistency for the

experimental setup that can expect increased robustness for perception algorithms that use the data. Finally, by incorporating VIL as a testbed for our methodology, we expect to lower the expenses for testing the robustness of perception algorithms by substituting real vehicle tests under harsh weather conditions.

2 Related Work

2.1 Adverse Weather and LiDAR Simulation

LiDAR sensor performance under adverse weather conditions, such as fog, rain, and snow, has been extensively studied using both real-world and simulated datasets (Bijelic et al., 2020; Carballo et al., 2020). While physics-based modeling (Hahner et al., 2021, 2022; Kilic et al., 2021) provides valuable insights, assuming rather uniform distributions of weather effects. Recent approaches leverage synthetic data augmentation to address this limitation. DigiWeather (Nikolov, 2024) and WeatherGen (Wu et al., 2025) have demonstrated effective synthetic augmentation methods, utilizing diffusion-based generative models to produce diverse and realistic LiDAR point clouds under varied weather conditions.

2.2 Neural Networks for LiDAR Data

Generative Adversarial Networks (Goodfellow et al., 2014) have proven capable of capturing complex distributions beyond image data (Wei et al., 2021; Zhu et al., 2017). Wasserstein GANs and their improved variants (WGAN-GP) effectively mitigate common training issues like mode collapse (Arjovsky et al., 2017; Gulrajani et al., 2017). Recently, conditional generative models have specifically addressed the challenge of simulating adverse weather conditions, producing realistic fog-affected LiDAR data (Lee et al., 2022; Zhang et al., 2024).

Long Short-Term Memory (LSTM) networks (Hochreiter & Schmidhuber, 1997) further enhance LiDAR data modeling by capturing temporal dependencies in sequential data. Combining GAN and LSTM architectures facilitates robust generation of spatiotemporally coherent data, improving the realism of synthetic fog conditions (Chen et al., 2023).

2.3 Robust 3D Object Detection

LiDAR-based 3D object detection algorithms have evolved significantly, from pioneering models such as PointNet (Qi et al., 2017) and VoxelNet (Zhou & Tuzel, 2018) to more recent developments like PV-RCNN++ (Shi et al., 2023), which combines voxel-level features and point-based processing to achieve enhanced detection accuracy.

Addressing robustness under adverse weather conditions, recent methods include AWARDistill (Liu et al., 2025), which employs knowledge distillation for adaptive robustness in 3D detection under varied weather. FogFusion (Zhang et al., 2025) integrates camera and LiDAR sensor data, specifically targeting improved detection in foggy environments through advanced sensor fusion strategies. Similarly, L4DR (Huang et al., 2025) proposes fusing LiDAR and 4D radar data, significantly improving detection reliability in adverse weather conditions.

2.4 LiDAR Simulation Platforms

LiDAR simulation is useful for validating sensor performance and algorithm robustness under controlled conditions. Physics-based simulations (Schick & Schindler, 2022; Teufel et al., 2022; Tremblay et al., 2021) provide direct representations of sensor–environment interactions, while data-driven approaches (Elmadawi et al., 2019; Jansen et al., 2022; Vacek et al., 2022) leverage machine learning to bypass complex physical modeling.

Simulation platforms such as IPG CarMaker enable realistic autonomous driving scenarios, integrating detailed LiDAR sensor models (Ahn et al., 2023; Kambampati, 2023). Open-source platforms Carla and LGSVL provide flexible virtual environments for rapid prototyping of diverse sensor configurations (Manivasagam et al., 2020; Rong et al., 2020).

3 DATA Acquisition

As authentic fog cannot be encountered on demand in the real world, we conducted the data acquisition sequence in a controlled environment using the Weather Simulation Facility (KATRI), a chamber with an internal width of approximately 20 m and length of 300 m located in Hwaseong-si, Gyeonggi-do, South Korea. The chamber generates artificial fog droplets and disperses them evenly, providing testing capabilities for sensor performance in adverse weather conditions.

A Velodyne VLP-32C LiDAR sensor was attached to a Hyundai Solati, which drove back and forth along the middle of the chamber under fog. As the vehicle is fully prepared for VIL experiments, it has its own in-house power supply and computation capacity for data acquisition and storage. Data from the LiDAR were accumulated using the ROS protocol and later converted to pcd format. By simplifying the experiment and data acquisition conditions, we have accumulated approximately 80,000 frames during 7 h of driving, ready to be fed into a neural network for training. The setup and acquired data are shown in Fig. 1.

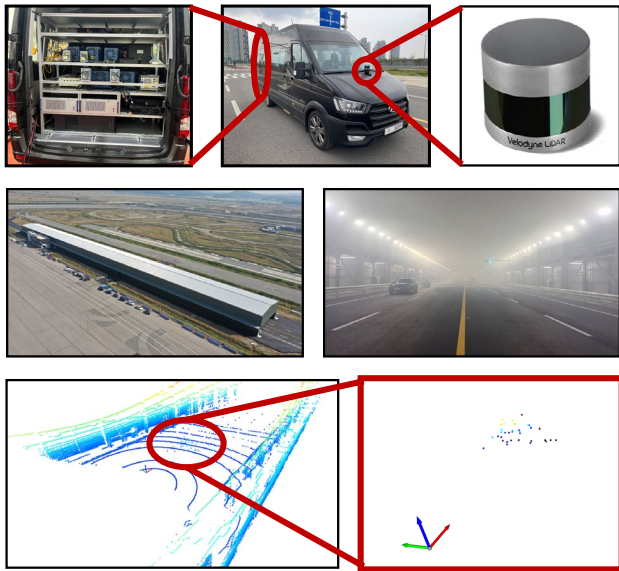


Fig. 1 Experimental setup and data acquisition. The top row shows the test vehicle with its hardware setup. The middle row shows the external façade and the internals of the Weather Simulation Facility. The bottom row shows a sample LiDAR point cloud acquired from the equipment and environment above, highlighting a point cloud cluster that represents fog

4 Fog Emulation

Although the point clouds for fog were separated from the experimentally acquired LiDAR data, numerical approximation of the distribution of fog for emulation is not a simple task. Fog as a phenomenon from a macroscopic perspective is simply a colloidal amalgamation of air and water droplets, but it is well accepted to assume that the emulation of fog as a group of individual droplets is complex and computationally taxing. Fog modeled as a body that allows partial transmission of light allows the model to be much lighter, but such physics-based modeling approaches can lead to overlooking nuanced characteristics.

In contrast, neural networks have demonstrated significant utility in identifying and incorporating these nuanced features that physics-based models may neglect, as they are trained on data closer to its raw form. In this study, we implement a generative neural network trained to generate data in the form of LiDAR point clouds that emulate the pattern of fog.

4.1 Background on Generative Adversarial Networks

Generative Adversarial Networks (GANs) and their variants are known for generating visually convincing images with relatively short inference times. In this work, we utilize these qualities to generate LiDAR detections affected by fog.

However, GANs are notoriously difficult to train, primarily due to challenges in achieving stable convergence between the generator and critic. These challenges include mode collapse, where the generator produces a limited variety of outputs. We employ a more stable and robust variant to overcome these difficulties: the Wasserstein GAN (WGAN). Unlike traditional GANs, which use the Jensen–Shannon divergence to distinguish real and generated data distributions, WGANs use the Earth Mover’s distance (also known as the Wasserstein distance) to estimate the distance between the real and generated data distributions. This training approach ensures improved convergence behavior and avoids pitfalls such as mode collapse, provided that the Lipschitz continuity condition is maintained through techniques like gradient penalization.

4.2 Background on Recurrent Neural Networks

To capture the temporal dynamics of LiDAR point clouds, we utilize a Long Short-Term Memory (LSTM) network, which is trained to include the temporal dependencies across consecutive frames of LiDAR data. We integrate the WGAN with LSTM by positioning the WGAN network between the fully connected layers of LSTM that produce each temporal output. This allows the network to learn both the spatial and temporal patterns associated with fog as detected by LiDAR.

4.3 Generative Neural Network Implementation and Training

In this study, we propose a generative model combining Wasserstein Generative Adversarial Networks (WGAN) with Long Short-Term Memory (LSTM) networks to simulate LiDAR data under foggy conditions realistically. The core motivation behind choosing a GAN-based method over diffusion models is the necessity for real-time generation, enabled by the GAN’s single forward pass, resulting in predictable inference latency. The architecture integrates LSTM to capture temporal dynamics inherent in sequential LiDAR point cloud data. Figure 2 provides a concise overview of the network structure.

4.4 Generator Structure

The generator comprises three main components: fully connected layers, residual convolutional blocks, and an integrated LSTM module.

4.4.1 Latent Representation and Conditioning

The input noise vector serves as a latent vector that passes through a fully connected layer which projects the latent

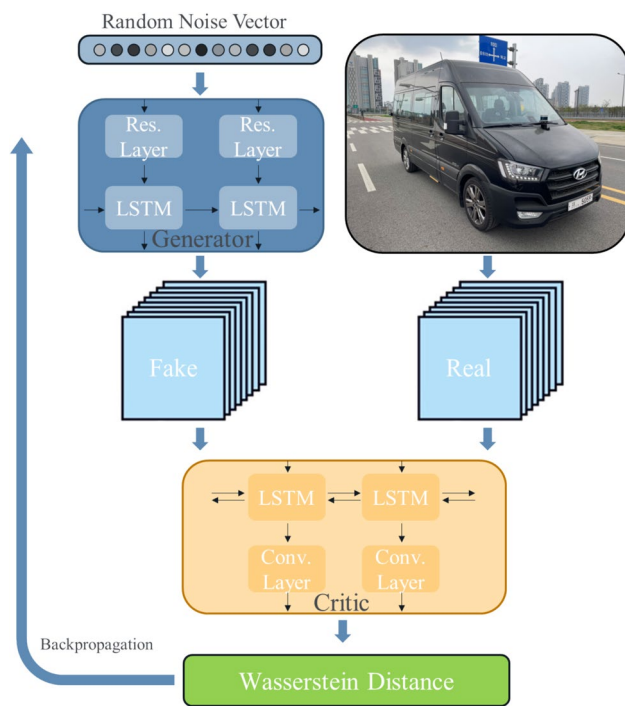


Fig. 2 High level representation of the neural network architecture. The critic and generator are trained simultaneously with opposite directives. Resulting Wasserstein distance is the loss of the system that is backpropagated

space into a high-dimensional tensor more suitable for successive convolutional operations.

4.4.2 Residual Blocks

This projection undergoes sequential processing through four residual blocks which resemble the original residual block used in the ResNet architecture. These blocks enhance the generative capacity by allowing the preservation of important features through skip connections, stabilizing gradient flows during training.

4.4.3 LSTM Integration for Temporal Modeling

The generator integrates an LSTM component through linear layers to maintain a hidden state and cell state capturing temporal dependencies across sequential frames. The hidden state and cell state evolves at each timestep based on the generator's output from previous timesteps, effectively encoding temporal coherence into the generated sequence.

4.4.4 Final Convolution and Activation

The final layer applies a convolutional operation followed by a hyperbolic tangent activation function to constrain the

generated data to a suitable range consistent with normalized LiDAR intensity data.

4.5 Critic Structure

The critic similarly utilizes convolutional layers combined with an LSTM module.

4.5.1 Convolutional Layers

Three convolutional layers progressively extract spatial features from the input LiDAR point cloud. These layers are interleaved with instance normalization and Leaky ReLU activation functions, stabilizing the network's response to varying input distributions.

4.5.2 Temporal Information via LSTM

Temporal consistency is introduced by an integrated LSTM structure involving convolutional recurrent layers followed by linear layers. The critic's hidden state and cell state maintains memory across timesteps, enhancing its capability to distinguish temporally coherent, realistic fog artifacts from synthetic ones lacking temporal realism.

4.5.3 Final Decision Layer

The flattened feature map from the final convolutional layer passes through a fully connected layer, producing a single scalar output used by the WGAN objective function.

4.6 Training Details

4.6.1 Gradient Penalty (WGAN-GP)

To overcome GANs' inherent instability, a Wasserstein loss with gradient penalty is employed. This method enforces Lipschitz continuity by penalizing gradients deviating from unity, ensuring stable convergence and robust training.

4.6.2 Optimization and Hyperparameters

Training is executed using the Adam optimizer with carefully selected hyperparameters (learning rate = $2e-5$, $\beta_1 = 0.05$, $\beta_2 = 0.95$). The critic is updated 4 times per single generator update, ensuring stable adversarial training.

4.6.3 Batch and Sequence Processing

Data batches are processed sequentially. Each timestep updates the hidden state of the LSTM, embedding temporal consistency in both generator and critic modules.

4.6.4 Training Dataset

Real foggy LiDAR data, collected from a controlled environment, are normalized to $[-1, 1]$ for numerical stability and better GAN performance. Corresponding synthetic fog samples generated by the network are compared against this normalized real data for loss calculation.

4.7 Objective Function

The governing objective function is identical to WGAN, as convolutional, residual, and LSTM layers are just building blocks for the WGAN-LSTM network

$$\tilde{x} = G(z) \quad (1)$$

$$\min_G \max_D \left[\mathbb{E}_{x \sim P_r} [D(x)] - \mathbb{E}_{\tilde{x} \sim P_g} [D(\tilde{x})] - \lambda \mathbb{E}_{\tilde{x} \sim P_{\tilde{x}}} [(\|\nabla_{\tilde{x}} D(\tilde{x})\|_2 - 1)^2] \right]. \quad (2)$$

The objective function represents the competitive relation between the generator and critic and the gradient penalty for Lipschitz continuity. The generator seeks to reduce the difference in outcome when the critic is fed with real and generated data, while the critic seeks the opposite.

4.8 Simulation of Generated Fog

Datasets for training are then generated with CarMaker, a vehicle dynamics simulation software developed by IPG Automotive GmbH. It also serves as a platform for diverse scenarios where various traffic objects are placed inside a virtual playground. The ego vehicle will then acquire LiDAR detection records from the surroundings with a LiDAR sensor simulated with ray tracing enabled on the GPU along with the annotations needed for training and validation.

By embedding the developed generator model inside the LiDAR sensor model interface, we can obtain fog-affected LiDAR detections from a software-defined scenario, which allows us to bypass the cost of a test drive in the real-world road environment. Figure 3 shows screenshots taken from the CarMaker software, while the simulation was running.

5 Results

5.1 Training Results

For quantitative analysis of generated fog distributions, we monitored key training metrics including critic loss, generator loss, and the Wasserstein distance between generated and real LiDAR fog images, as plotted in Fig. 4. The training progression showed a stable convergence pattern: the

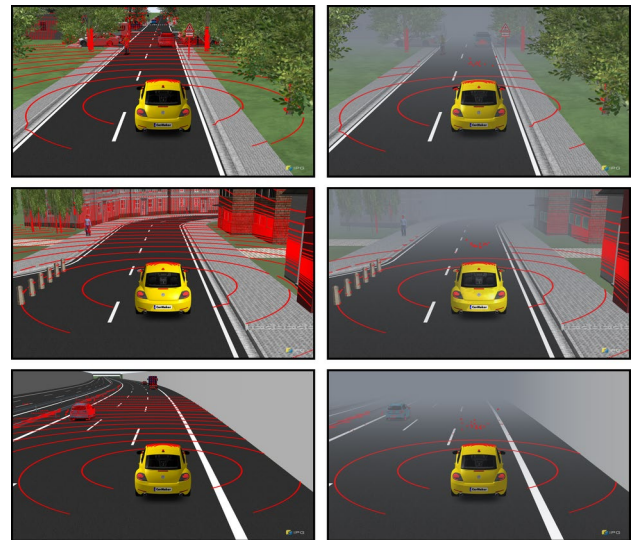


Fig. 3 Comparison of CarMaker simulation for LiDAR detection points in clear and foggy weather in different scenarios. Note that the gray fog effect is just a visual effect without any influence on the LiDAR detection points from the virtual sensor or the generator network

critic loss consistently stabilized near zero, indicative of effective critic training and balance against the generator. Correspondingly, the generator loss decreased progressively, suggesting improvements in generating realistic fog patterns. Finally, the measured Wasserstein distance between real and generated data rapidly decreased during initial epochs and stabilized at a low value, signifying the generated fog distributions closely approximated the real fog data. These metrics collectively indicate the effectiveness and accuracy of the proposed generative approach in synthesizing realistic LiDAR fog scenarios.

Qualitative analysis can be done by examining the 2D footprint of the LiDAR detections, as shown in Fig. 5. It is high impossible for the naked eye to tell whether a frame is from the batch of real images, or from the batch of generated images.

5.2 Perception Results

In the perspective of the fact that LiDAR sensor data are not strictly designed or intended for human interpretation, using 3D object detection methods following the generation of simulated fog is one method for validation. As LiDAR sensor data are typically processed by another parameterized model to extract meaningful information, such as deriving bounding box coordinates from abstract point clouds, feedback from the perception model—the consumer of the LiDAR data—is as valuable as feedback from any other sources.

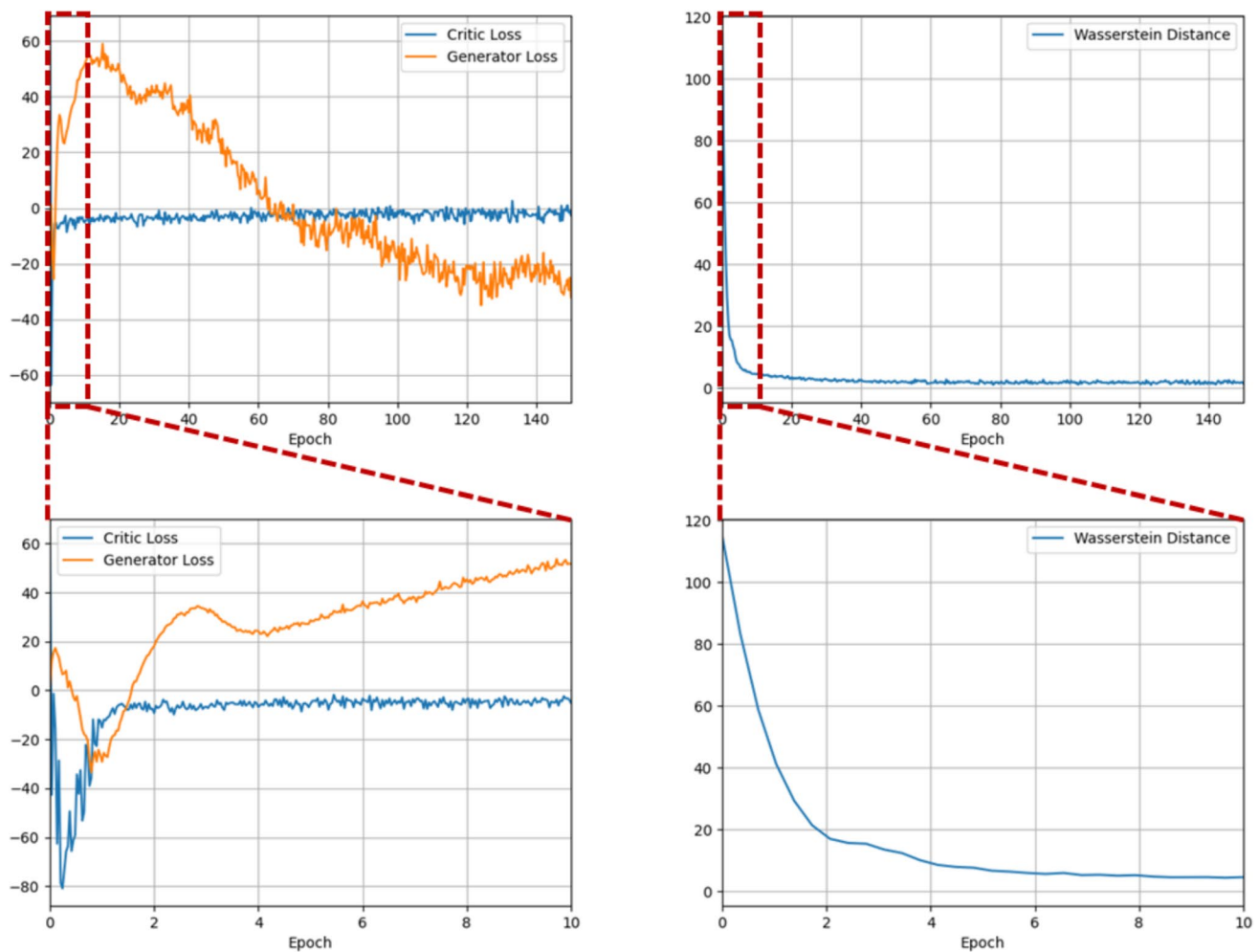


Fig. 4 Training progress of the proposed generative model. Plots on the left depict the evolution of generator and critic losses over epochs, showing stabilization and convergence after initial fluctuations. Plots

on the right show the Wasserstein distance between the real and generated point cloud data, which is different from the critic loss

For this purpose, preexisting 3D object detection models were forked from OpenPCDet (2024). Their Github repository includes implementations of algorithms, such as PV-RCNN (Shi et al., 2020a, 2020b), PointPillars (Lang et al., 2019), CenterPoint (Yin et al., 2021), Part-A² (Shi et al., 2020a, 2020b), and SECOND (Yan et al., 2018). Using the default configuration for each method with minor adjustments for custom data input, we train these models for 80 epochs. For each available perception method, we train one model on data without fog and another model on data with fog.

The metric we employ for quantification is mean average precision (mAP), a common metric for evaluating perception algorithms. By comparing the two models trained on different datasets and their performance when fed with foggy data, we claim that our approach involves considerable improvements. Table 1 shows larger and unanimous improvements in detecting vehicles. The training dataset that performed better

on foggy evaluation datasets is marked bold. Table 2 shows a more thorough investigation including results for different object classes. For pedestrians and cyclists, improvements vary by model. The training dataset with less difference between the evaluation results on clear and foggy datasets is marked bold.

6 Conclusion

In this paper, we introduced a novel methodology for simulating LiDAR data under foggy conditions using a Wasserstein Generative Adversarial Network combined with Long Short-Term Memory (WGAN-LSTM). By training the network on real fog-affected LiDAR point clouds collected in a controlled environment, we generated realistic fog artifacts that emulate both spatial and temporal characteristics of actual fog. Integrating this generative model into a virtual

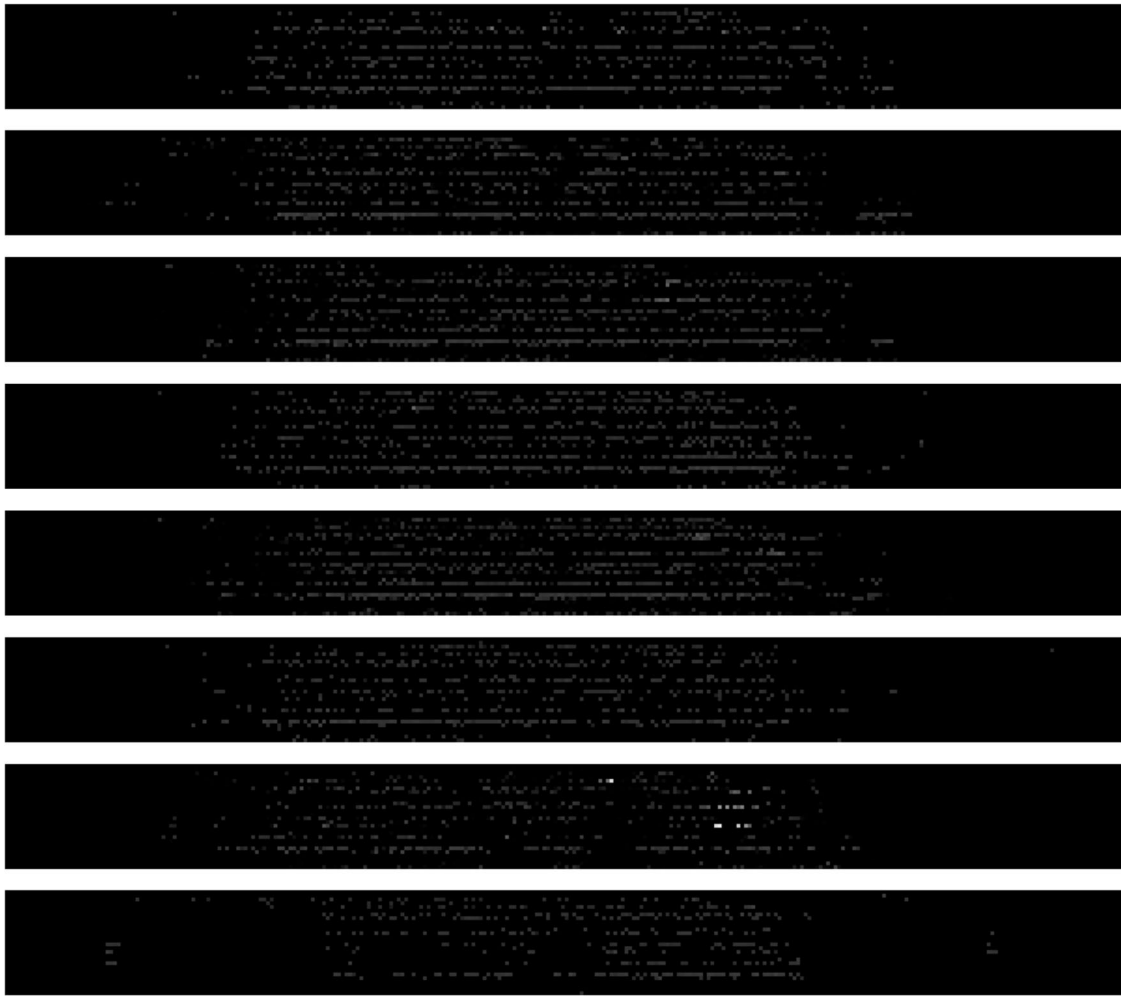


Fig. 5 LiDAR fog detections mapped onto 2D array. Images 1, 4, 6, 8 from the top are real images acquired from the Velodyne VLP-32C. Images 2, 3, 5, 7 are generated from random noise vectors forwarded through the generator neural network

simulation environment allowed us to produce fog-affected LiDAR data in real time, significantly reducing the need for costly and time-consuming real-world data collection under adverse weather conditions.

Our approach was validated through extensive experiments using various state-of-the-art 3D object detection algorithms. The results demonstrated a substantial improvement in mean average precision (mAP) when the models were trained on datasets augmented with our generated foggy LiDAR data. Specifically, models trained on foggy data outperformed those trained solely on clear-weather data when evaluated under foggy conditions, highlighting

the effectiveness of our method in enhancing the robustness of perception algorithms (Fig. 6).

In conclusion, our work offers an efficient and reusable solution to the challenge of developing and validating robust 3D object detection methods. By leveraging generative neural networks to simulate realistic fog effects on LiDAR data, we facilitate the production of diverse training datasets that help enhance the performance and resilience of perception algorithms. This approach not only accelerates the development process but also contributes to the safety and reliability of autonomous vehicles operating in diverse environmental conditions.

Table 1 Comparison of mAP (bounding box) of 3D object detection methods trained on clear-weather data and foggy weather data during a foggy weather simulation

Method	Training dataset	Evaluation dataset	Car mAP	Car mAP@R40
Second	Clear	Fog	56.0924	55.0098
	Fog		70.2672	71.0497
	Δ		– 14.1748	– 16.0399
Second-iou	Clear	Fog	36.3325	29.9884
	Fog		71.5005	68.9636
	Δ		– 35.168	– 38.9752
CenterPoint	Clear	Fog	27.2117	24.8909
	Fog		71.9785	71.7633
	Δ		– 44.7668	– 46.8724
PartA ²	Clear	Fog	35.6581	29.4278
	Fog		45.3247	39.9154
	Δ		– 9.6666	– 10.4876
PointPillar	Clear	Fog	45.2288	42.277
	Fog		70.7521	72.5949
	Δ		– 25.5233	– 30.3179
PV-RCNN	Clear	Fog	27.0299	19.8784
	Fog		54.2466	56.7616
	Δ		– 27.2167	– 36.8832

Table 2 Complete comparison of mAP (bounding box) encompassing all combinations of 3D object detection methods, training dataset weather, testing dataset weather, and class of detected objects

Method	Training dataset	Evaluation dataset	Car mAP	Car mAP@R40	Pedestrian mAP	Pedestrian mAP@R40	Cyclist mAP	Cyclist mAP@R40
Second	Clear	Clear	76.6489	78.6849	39.1925	37.1358	16.8296	12.1585
		Fog	56.0924	55.0098	40.6322	39.191	5.4485	3.3542
		Δ	20.5565	23.6751	− 1.4397	− 2.0552	11.3811	8.8043
	Fog	Clear	65.3462	65.554	40.995	39.4285	20.4048	15.983
		Fog	70.2672	71.0497	36.094	32.1995	12.378	6.427
		Δ	− 4.921	− 5.4957	4.901	7.229	8.0268	9.556
Second-iou	Clear	Clear	63.5152	64.7649	41.9454	42.8854	16.8182	11.0625
		Fog	36.3325	29.9884	31.9544	28.0493	6.8182	1.875
		Δ	27.1827	34.7765	9.991	14.8361	10	9.1875
	Fog	Clear	80.3535	81.5534	27.0109	26.5889	9.0909	1.4286
		Fog	71.5005	68.9636	26.4059	21.7182	9.1209	3.5
		Δ	8.853	12.5898	0.605	4.8707	− 0.03	− 2.0714
CenterPoint	Clear	Clear	63.5502	66.888	51.9113	50.1885	53.2915	51.5812
		Fog	27.2117	24.8909	49.467	49.1449	18.1818	13.9236
		Δ	36.3385	41.9971	2.4443	1.0436	35.1097	37.6576
	Fog	Clear	86.0334	86.8154	51.6494	49.2968	38.7781	36.4758
		Fog	71.9785	71.7633	51.5897	49.3891	47.8095	45.8241
		Δ	14.0549	15.0521	0.0597	-0.0923	− 9.0314	− 9.3483
PartA ²	Clear	Clear	63.0455	62.0393	27.1128	22.3407	14.7727	11.3583
		Fog	35.6581	29.4278	17.2551	15.9886	3.0303	1.7613
		Δ	27.3874	32.6115	9.8577	6.3521	11.7424	9.597
	Fog	Clear	66.0057	65.3862	25.5657	20.5741	4.5455	2.4369
		Fog	45.3247	39.9154	17.5423	11.9487	9.0909	1.4286
		Δ	20.681	25.4708	8.0234	8.6254	− 4.5454	1.0083
PointPillar	Clear	Clear	71.479	69.1016	51.8433	51.4682	49.1642	48.8608
		Fog	45.2288	42.277	26.346	25.5616	16.1616	8.8889
		Δ	26.2502	26.8246	25.4973	25.9066	33.0026	39.9719
	Fog	Clear	78.8391	79.4951	73.1586	73.588	25.4417	24.5695
		Fog	70.7521	72.5949	35.5681	36.2128	15.5844	10.0092
		Δ	8.087	6.9002	37.5905	37.3752	9.8573	14.5603
PV-RCNN	Clear	Clear	63.4558	64.6548	51.2569	47.1778	31.4935	26.6716
		Fog	27.0299	19.8784	31.9045	31.8216	21.1081	15.8044
		Δ	36.4259	44.7764	19.3524	15.3562	10.3854	10.8672
	Fog	Clear	62.2864	65.5262	26.9879	26.5118	16.9697	9.4583
		Fog	54.2466	56.7616	26.5283	21.9741	9.0909	4.6429
		Δ	8.0398	8.7646	0.4596	4.5377	7.8788	4.8154

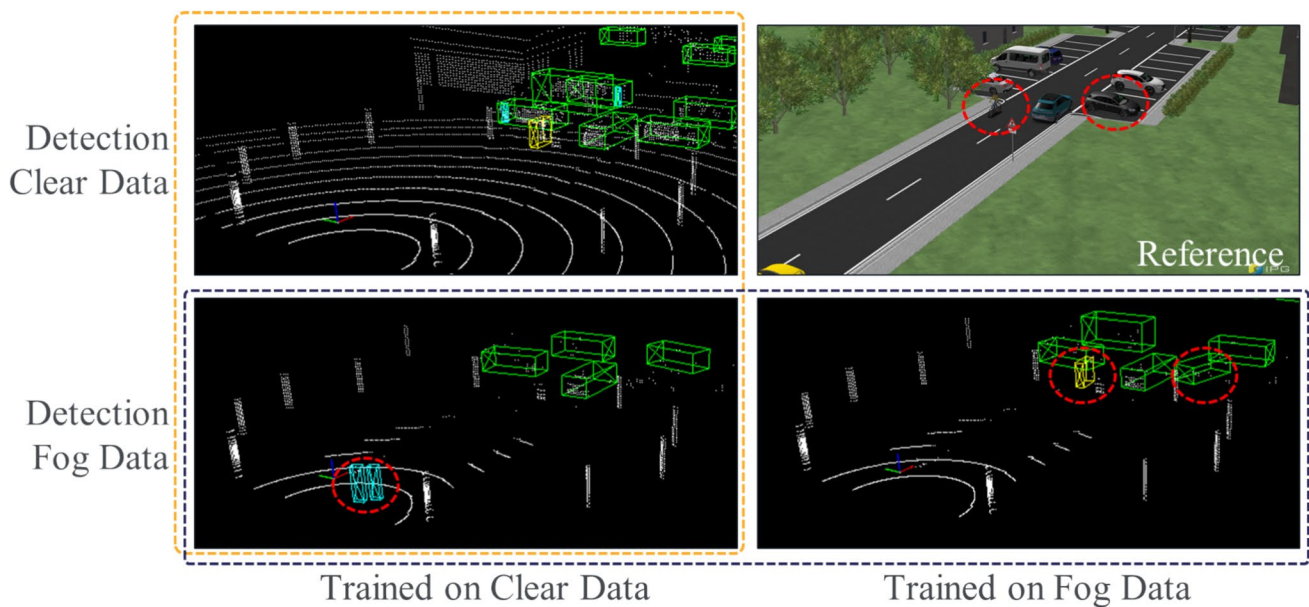


Fig. 6 A visual example of how the performance of a 3D object detection network (PointPillars) trained with clear-weather data is degraded in a foggy simulation and how the same network trained on foggy weather data successfully identifies items that were unidentified by the former

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Data availability Data supporting this study are available from the corresponding author on request.

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